

Curriculum Vitae

Nikita Kolosov, M.Sc.

Current position:

2023 (October) - present Graduate Research Student
The Ohio State University, Columbus, OH, USA;
The Steve and Cindy Rasmussen Institute for Genomic
Medicine, Nationwide Children's Hospital, Columbus, OH,
USA

Additional affiliations:

2020-present Associated Researcher
Broad Institute, Cambridge, MA, USA

2020-present Visiting Researcher
Institute for Molecular Medicine (FIMM),
Helsinki, Finland

Past positions:

2023-2023 Senior Research Associate
Nationwide Children's Hospital, Columbus, OH, USA

2022-2023 Research Consultant
Analytic and Translational Genetics Unit
Massachusetts General Hospital, Boston, MA, USA

2020-2022 Staff Scientist
World-Class Research Center for Personalized Medicine
Almazov National Research Centre, St. Petersburg, Russia

2019-2022 Junior Programmer
Center for Genomic Diversity
ITMO University, St. Petersburg, Russia

Research expertise:

My research focuses on identifying genetic and phenotypic susceptibilities to complex traits. Specifically, I develop computational tools, pipelines, and statistical models to predict disease risk or age of disease onset. I have extensive experience analyzing large-scale biobank data, including both genotypic and phenotypic (static and longitudinal) datasets from resources such as FinnGen and UK Biobank (UKBB).

Education:

- 2023-present PhD student at The Ohio State University, OH, USA
Program: Biomedical Sciences Graduate Program
Specialization: Biomedical informatics
- 2018-2020 M.Sc., ITMO University, Saint-Petersburg, Russia
Department: Information Technology and Programming
Specialization: Bioinformatics and Systems Biology
- 2014-2018 B.Sc., Saint-Petersburg State University, Saint-Petersburg, Russia
Department: Biological Sciences
Specialization: Developmental Biology

Teaching:

- 2022 Lecturer
"Winter School of Immunology"
Université Côte d'Azur (Nice, France)
- 2021-2024 Lecturer
"Bioinformatic Methods of Genomic Data Analysis"
Almazov National Medical Research Centre
(Saint-Petersburg, Russia)
- 2019-2020 Teaching Assistant
"Data Processing and Analysis" course
ITMO University (Saint-Petersburg, Russia)

Awards:

- 2024 - Outstanding Graduate Student for the Abigail Wexner Research Institute at Nationwide Children's Hospital (NCH)
- 2024 - Trainee Research Excellence Finalist Award, American Society of Human Genetics (ASHG)
- 2023 - Abstract of distinction, Nationwide Children's Hospital research retreat
Nov. 15, 2023
- 2020-2021 - JetBrains research stipend

Skills/Competence:

- Computational ML, Python, R, bash, git, docker
- Bioinformatic GWAS, PRS, RNA-seq, NGS

Conferences, hackathons and talks:

- 1) Platform talk at American Society of Human Genetics (ASHG) 2024 conference (Denver CO, USA); "PGS Browser: a comprehensive analysis of 3,168 polygenic score models across 400,000 Finns".
- 2) Poster presentation - American Society of Human Genetics (ASHG) 2023 conference (Washington DC, USA); "FinnGen PGS Browser: a platform for polygenic-risk score analyses and interpretation from a cohort of 400,000 Finns".
- 3) Platform talk, 4th Saint Petersburg Medical Innovation Forum, 2021 (St. Petersburg, Russia).
- 4) Winner of evolutionary bioinformatics hackathon (EBH) 2018 (St. Petersburg, Russia), resulted in a publication: "Retroelement-Linked H3K4me1 Histone Tags Uncover Regulatory Evolution Trends of Gene Enhancers and Feature Quickly Evolving Molecular Processes in Human Physiology".

Publications:

- 1) Dmitrii Usoltsev, **Nikita Kolosov**, Oxana Rotar, Alexander Loboda, Maria Boyarinova, Ekaterina Moguchaya, Ekaterina Kolesova, Anastasia Erina, Kristina Tolkunova, Valeriia Rezapova, Olesya Melnik, Olga Freylikhman, Nadezhda Paskar, Asiat Alieva, Elena Baranova, Elena Bazhenova, Olga Beliaeva, Elena Vasilyeva, Sofia Kibkalo, Rostislav Skitchenko, Alina Babenko, Alexey Sergushichev, Alena Dushina, Ekaterina Lopina, Irina Basyrova, Roman Libis, Dmitrii Duplyakov, Natalya Cherepanova, Kati Donner, Paivi Laiho, Anna Kostareva, Alexandra Konradi, Evgeny Shlyakhto, Aarno Palotie, Mark J. Daly, Mykyta Artomov (2023). "Complex trait susceptibilities and population diversity in a sample of 4,145 Russians". *Nature Commun* **15**, 6212 (2024). <https://doi.org/10.1038/s41467-024-50304-1>
- 2) Mary Pat Reeve, Mari Vehviläinen, Shuang Luo, Jarmo Ritari, Juha Karjalainen, Javier Gracia-Tabuenca, Juha Mehtonen, Shanmukha Sampath Padmanabhuni, **Nikita Kolosov**, Mykyta Artomov, Harri Siirtola, Hanna M. Ollila, FinnGen, Daniel Graham, Jukka Partanen, Ramnik J. Xavier, Mark J. Daly, Samuli Ripatti, Tuula Salo, Maria Siponen (2024). "Oral and non-oral lichen planus show genetic heterogeneity and differential risk for autoimmune disease and oral cancer". *The American Journal of Human Genetics*, Volume 111, Issue 6, 1047 - 1060
- 3) Daria Pinakhina, Danat Yermakovich, Ekaterina Vergasova, Evgeny Kasyanov, Grigory Rukavishnikov, Valeriia Rezapova, **Nikita Kolosov**, Alexey Sergushichev, Iaroslav Popov, Elena Kovalenko, Anna Ilinskaya, Anna Kim, Nikolay Plotnikov, Valery Ilinsky, Nikholay Neznanov, Galina Mazo, Alexander Kibitov, Alexander Rakitko, Mykyta Artomov (2023) "GWAS of depression in 4,520 individuals from the Russian population highlights the role of *MAGI2* (*S-SCAM*) in the gut-brain

axis". *Frontiers in Genetics*. 13:972196. doi: 10.3389/fgene.2022.972196

- 4) **Nikita Kolosov**, Valeriia Rezapova, Oxana Rotar, Alexander Loboda, Olga Freylikhman, Olesya Melnik, Alexey Sergushichev, Christine Stevens, Trudy Voortman, Anna Kostareva, Alexandra Konradi, Mark J. Daly, Mykyta Artomov (2022). "Genotype Imputation and Polygenic Score Estimation in Northwestern Russian Population." *PloS One* 17 (6): e0269434.
- 5) **Nikita Kolosov**, Mark J. Daly, Mykyta Artomov (2021). "Prioritization of Disease Genes from GWAS Using Ensemble-Based Positive-Unlabeled Learning." *European Journal of Human Genetics: EJHG* 29 (10): 1527–35.
- 6) Daniil Nikitin, **Nikita Kolosov**, Anastasiia Murzina, Karina Pats, Anton Zamyatin, Victor Tkachev, Maxim Sorokin, Philippe Kopylov, and Anton Buzdin. 2019. "Retroelement-Linked H3K4me1 Histone Tags Uncover Regulatory Evolution Trends of Gene Enhancers and Feature Quickly Evolving Molecular Processes in Human Physiology" *Cells* 8, no. 10: 1219. <https://doi.org/10.3390/cells8101219>

Publication in-review:

- 1) Mary Reeve, Masahiro Kanai, Daniel Graham, Juha Karjalainen, Shuang Luo, **Nikita Kolosov** et al. Autoimmune hypothyroidism GWAS reveals independent autoimmune and thyroid-specific contributions and an inverse relation with cancer risk, 08 July 2024, PREPRINT (Version 1) available at *Research Square*; <https://doi.org/10.21203/rs.3.rs-4626646/v1>.
- 2) **Nikita Kolosov**, Mary P. Reeve, Pietro Della Briotta Parolo, Mitja I. Kurki, Vincent Llorens, FinnGen, Timo Petteri Sipilä, Adam Herman, Ivan Molotkov, Mervi Aavikko, Samuli Ripatti, Aarno Palotie, Mark J. Daly, Mykyta Artomov (2025). "PGS Browser: a public platform for personalized polygenic score interpretation" *medRxiv* 2025.07.03.25330851; <https://doi.org/10.1101/2025.07.03.25330851>

Profiles:

ResearchGate: <https://www.researchgate.net/profile/Nikita-Kolosov>

References:

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